“The purpose of visualization is

insight, not pictures.”

Data Visualization and Storytelling

ECBS5209 · CEU · Fall 2026
Ulrich Wohak · 4 November 2026

Welcome to the course

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MSc Business Analytics · 1 US credit / 2 ECTS

Six Wednesday sessions · 17:40–19:20

Bring your laptop and your existing Python and pandas skills.

How we will work in class

Discuss a question and inspect the evidence.

Build Python examples, then Power BI charts and dashboards.

Inspect a chart and decide what you would change.

Explain your choices and check the result.

The code may run perfectly while the chart communicates badly.

Our six sessions

Date	Focus
4 Nov	**Python chart tour:** line, bar, scatter, histogram, box, violin and strip plots; labels, notes and export.
11 Nov	**Comparisons and uncertainty:** histograms, box plots and confidence-interval plots, including a supplied event-study example.
18 Nov	**Mapping in Python:** geographical joins, choropleths, proportional symbols, projections and meaningful denominators.
25 Nov	**Power BI foundations:** interface, importing data, field types, aggregation and basic charts.
2 Dec	**Power BI comparisons and filtering:** tables, matrices, cards/KPIs, filters and slicers.
9 Dec	**Power BI dashboard building:** layout, interactions, storytelling and checking filtered results.

Wednesdays, 17:40–19:20 · 2026

Assessment

25% Three individual figures with notes

Equal weight within the combined 25%

75% Final Python data story

In-class exercises: practice and feedback

Minimum overall mark to pass: 50%

Submission dates and routes: to be announced

Three individual figures · 25% in total

Three submissions with equal weight.

For each, I supply the data and the question.

You design a figure in Python that answers it.

Submit only the figure with its accompanying notes.

Data, questions and deadlines will be announced later.

The final Python data story · 75%

One coherent question and a specific audience.

A sequence of 3–5 figures that work together.

Reproducible code, data provenance and a written narrative.

Work individually or in a team of up to three.

A strong static story is enough. Interactivity is optional.

Where to find the material

github.com/ulrichwohak/Python-Dataviz

One teaching notebook and one exercise notebook per lecture

Python: pandas, matplotlib and seaborn · environment: uv

No required textbook purchase. Readings will be identified in class.

Explore datasets: github.com/toddrjones/datasets

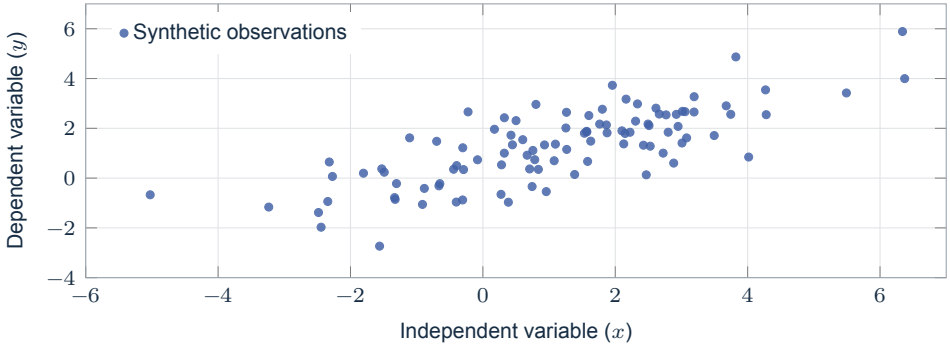
What are we trying to make small?

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{\beta_0, \beta_1}{\arg\min} \sum_{i=1}^{n} \left[y_i - (\beta_0 + \beta_1 x_i) \right]^2$$

$$\widehat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\arg\min} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^{\mathsf{T}} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

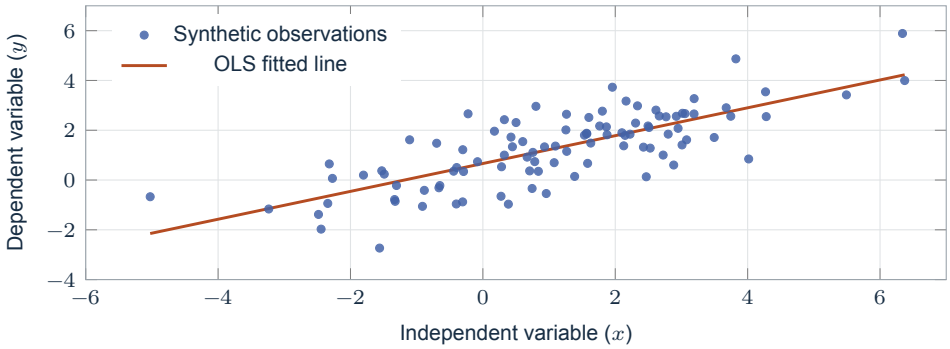
The equation gives precision. Can a picture give intuition?

A relationship with variation



100 synthetic observations

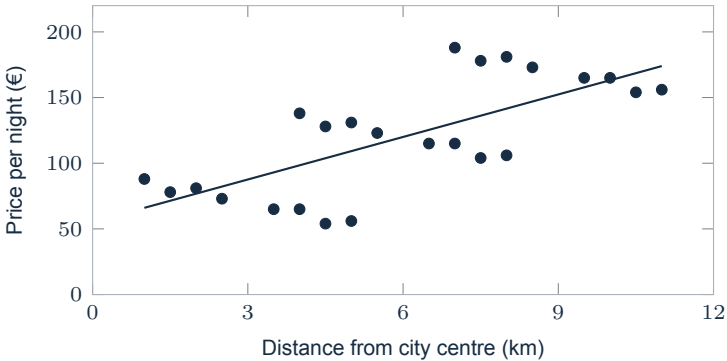
The OLS fitted line



100 synthetic observations. The line minimises the sum of squared vertical residuals.

“We understand *because* we see.” — Alberto Cairo, p. xv

What would you conclude from this chart?

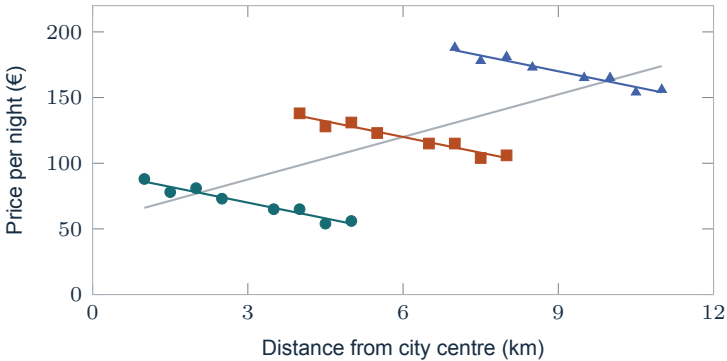


Across all hotels: price rises with distance.

What information might be missing?

Synthetic illustration · 24 hotels · three fictional countries · association, not causation

The same points tell a different story

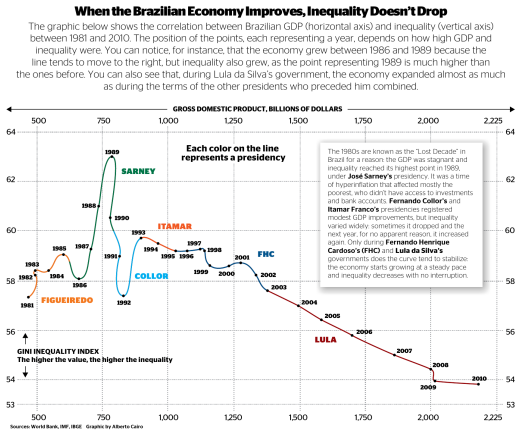


Within each country: price falls with distance.

- Country A
- Country B
- Country C

Synthetic illustration · same points and scales · association, not causation

What is this figure trying to communicate?

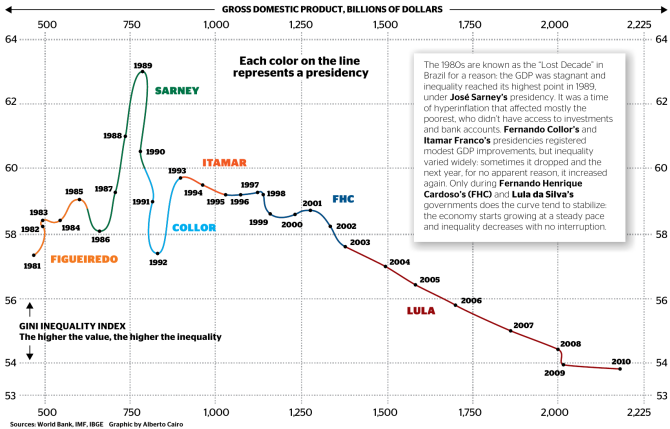


Cairo / Época · *The Functional Art*, Figure 3, p. xix

Describe the takeaway in one sentence.

Would another reader choose the same message?

Which details serve the intended purpose?



Cairo / Época · Figure 3, p. xix · same figure, enlarged chart detail

A focused finding?

Historical exploration?

Keep, remove, or separate?

Each figure needs a purpose and context

Who is reading, and what are they trying to understand?

What should this figure reveal, support or let them explore?

What would become misleading if we left it out?

Keep necessary units, sample, uncertainty and source notes.

Simplify the presentation while preserving the evidence.

Visualisation examples

Examples of Bad Visualisations

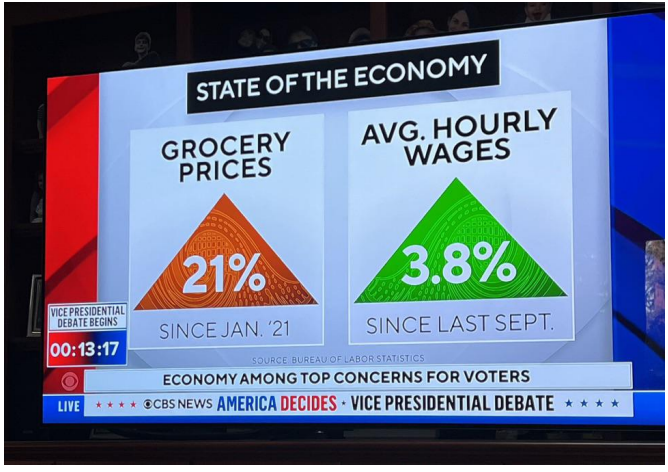
viz.wtf

reddit.com/r/dataisugly

Examples of Good Visualisations

reddit.com/r/dataisbeautiful

Prices and wages

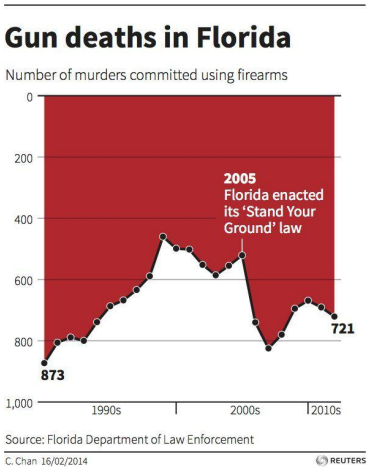


What comparison does this picture invite?

What would you check before accepting it?

Instructor-supplied photo showing CBS News and a BLS credit. Figures not independently checked.

Reading a time series



What happened after 2005?

Does your first impression agree with the scale?

Instructor-supplied image credited to Reuters / C. Chan, 16 February 2014.

A pool and three skyscrapers

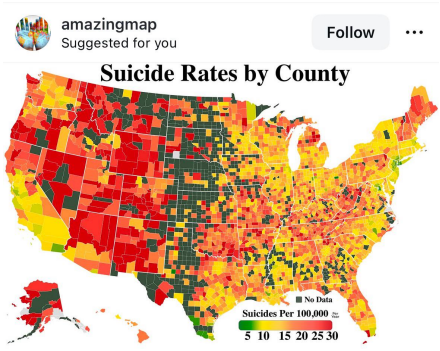


What does “bigger” mean here?

Do the labels and picture support the same comparison?

Instructor-supplied social-media screenshot. Original post and measurements not verified.

Reading a county map



What do the colours tell you?

What else would you need to interpret this map?

Instructor-supplied screenshot labelled amazingmap. Underlying source and data period not shown.

Today: a chart tour in Python

US birth counts: what changes across weekdays?

Line, bar and scatter plots for concrete questions.

Histograms, box plots, violins and individual data points.

Add labels and figure notes, then export a figure.

Final 20 minutes: redesign the supplied chart and check the values.

Keep the question visible while we write the code.

Backup · setup and Power BI access

Python: from the repository root

```
uv sync
uv run python scripts/fetch_data.py
uv run jupyter lab
```

First data download: about 330 MB. Later runs use verified local caches.

Before Session 4: Power BI access will be confirmed.

Windows / CEU access route: to be announced.

No purchase is required for this course.